

Roll No.....

ODD END SEMESTER EXAMINATION May-June - 2025

Name of the Course: B. Tech (EE)

Name of the Paper: Robotics

Time: 3 Hours

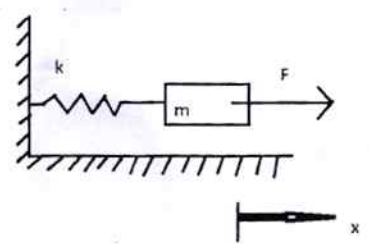
Semester: 8th

Paper Code: TOE 805

Maximum Marks: 100

Note:

- (i) All questions are compulsory
- (ii) Answer any two sub questions among a, b & c in each main question
- (iii) Total marks in each main question are twenty
- (iv) Each question carries 10 marks

Q1	(10 marks)	
(a)	Explain different types of sensors used in robot functioning and their applications.	
(b)	In a robotic setup, a camera is attached to the fifth link of a robot with six degrees of freedom. The camera observes an object and determines its frame relative to the camera's frame. Using the following information, determine the necessary motion the end effector has to make to get to the object: ${}^5T_{\text{cam}} = \begin{bmatrix} 0 & 0 & -1 & 3 \\ 0 & -1 & 0 & 0 \\ -1 & 0 & 0 & 5 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad {}^5T_H = \begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ ${}^{\text{cam}}T_{\text{obj}} = \begin{bmatrix} 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 4 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad {}^HT_E = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	CO1, CO5
(c)	Define different types of robotic configurations for positioning the hand of a robot.	
Q2	(10 marks)	
(a)	A human hand has seven degrees of freedom. Justify.	
(b)	Derive the force acceleration relationship for the one degree of freedom system for the following figure, using both the Lagrangian mechanics as well as the Newtonian mechanics. Assume that the wheels have negligible inertia. 	CO4, CO2, CO1
(c)	How the differential motion of the joint of a robot relates to differential motion of hand frame of robot?	
Q3	(10 marks)	CO3,

(a)	A point $P(7,3,2)^T$ is attached to a frame $(\bar{n}, \bar{o}, \bar{a})$ and is subjected to the below mentioned transformations. Find the coordinates of the point relative to the reference frame after the transformations. (1) Rotation of 90° about z-axis (2) Followed by a rotation of 90° about the y axis (3) Followed by a translation of $[4,-3,7]$	CO6
(b)	What do you mean by RPY angles. Define with suitable diagrams.	
(c)	It is desired to place the origin of the hand of a spherical robot at $[3 \ 4 \ 7]^T$. Calculate the joint variables of the robot.	
Q4	(10 marks)	CO3, CO2, CO6
(a)	Explain the anatomy of an industrial robot with neat diagrams.	
(b)	Explain different motions of a frame in brief. Find the total differential transformation caused by small rotations about the three axes of $\delta x=0.1$, $\delta y=0.05$ and $\delta z=0.02$ radians.	
(c)	Describe the historical evolution of robotics with key milestones.	
Q5	(10 marks)	CO2, CO4, CO1
(a)	It is desired to have the first joint of a six axis robot go from initial angle of 30° to a final angle of 75° in 5 seconds. Using a third order polynomial, calculate the joint angle at 1,2,3,4 seconds.	
(b)	Name various types of electric motors used in the functioning of robots. Explain any one of those.	
(c)	Discuss various types of joints in robots in detail.	